

Assignment6 – Learning

Given: May 27 Due: June 1

Problem 6.1 (Decision Trees)

Consider a word W chosen uniformly from $\{bad, bed, bend, pend, pad, ped\}$. You are allowed to ask the following questions about W :

- A length of the word
 - B first letter of the word
 - C second letter of the word
 - D last letter of the word
1. Show that there is no *decision tree* for W of depth 2.
 2. Draw the *decision tree* for W that arises from asking the questions in the order A,B,C,D. (Do not ask additional questions if the word can already be identified.)
 3. Calculate the *information gain* for all 4 questions.
 4. Which question would the *information gain algorithm* ask first?

We write $Q \succ A$ if the set Q of attributes determines the attribute A , i.e., examples that agree on all attributes in Q also agree on A .

5. Give all minimal sets $Q \subseteq \{A, B, C, D\}$ of questions for which the *determination* $Q \succ W$ holds?
6. Give two words such that removing them from the choices for W makes the *determination* $\{A, B\} \succ W$ hold.

Problem 6.2 (Decision Tree Learning in Python)

Implement the *Decision Tree Learning algorithm (DTL)* in Python using the files at <https://kwarc.info/teaching/AI/resources/AI2/dtl>.

Problem 6.3 (Loss)

Our goal is to find a *linear approximation* $h_a(x) = ax$ for the *sequence of squares* 0, 1, 4, 9, 16 of the *numbers* 0, 1, 2, 3, 4.

1. *Model* this situation as an *inductive learning problem*.
2. Assuming all 5 possible *examples* are *equally probable*, compute the *generalization loss* using the *squared error loss function*.
3. Explain in about 2 *sentences* how you would continue to eventually determine that $h^*(x) = 10x/3$.
4. What is the *error rate* of h^* ?

Problem 6.4 (Overfitting)

Explain what *overfitting* means and why we want to avoid it.

Problem 6.5 (Competition (due September 15))

In this competition, you will implement an agent that explores the FAULumpus world. You will receive up to 2 percentage points of additional bonus for your agents.

Submission deadline for your agents is September 15. All further details will be posted in the respective matrix channel.